Introduction to High-Performance Computing

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CAVEAT & instructions

- ✓ This is only introductory stuff
- ✓ There's much more under the hood
- ✓ These 6 lessons (three hour each) can only help you introduce some basic ideas
- ✓ More experience is needed (and so many mistakes) to manage HPC
- ✓ Almost all examples are in Fortran but the reasons behind performance are (almost) language independent (I'm a boomers)
- ✓ Please write to g.amaticode@gmail.com and alessandro.ceci@gssi.it to have your email
- ✓ material downloadable, after some fix, from https://github.com/gamati01/HPCLessons

Course structure/1

These are the HPC topics covered in this course:

- ✓ HPC description
- ✓ Memory subsystem
- ✓ Floating point units and structure
- ✓ Algorithm vs. Implementation
- ✓ Compiler
- ✓ Parallel paradigm for CPU (introduction)
- ✓ Parallel paradigm per GPU (introduction)

Course structure/2

Topics not covered here:

- ✓ OS
- ✓ Virtual Memory
- ✓ Parallel I/O
- ✓ ….

Example/Exercises

Some examples/exercises that will be presented/described:

- ✓ Matrix-Matrix Multiplication
- ✓ Matrix-Matrix Multiplication
- ✓ Matrix-Matrix Multiplication (again…)
- ✓ Laplace equation
- ✓ Travel Salesman problem
- ✓ Sieve of eratosthenes
- ✓ ….

You are not obliged to make exercise but is strongly suggested to play with them

Pre-requisites

What is requested:

- ✓ A Laptop/PC to perform some tests/exercises
- ✓ A basic knowledge of a programming language
  - Fortran/C/matlab/python….
- ✓ Enough Patience!

Who we are?

Just to know where we come from…

- ✓ G. Amati
- ✓ A. Ceci

References

Some books….

- ✓ Charles Severance; Kevin Dowd “High Performance Computing”, O’Reilly, ISBN 13:9781565923126
- ✓ John L. Hennessy, David A. Patterson , “Computer Architecture: A Quantitative Approach” Morgan Kaufmann; ISBN-10 : 0128119055
- ✓ John L. Hennessy, David A. Patterson , “Computer Organization and Design: The Hardware/Software Interface”, Morgan Kaufmann;
- ✓ D. Goldberg, ”What Every Computer Scientist Should Know About Floating-Point Arithmetic”
- ✓ U. Drepper: “What Every Programmer Should Know About Memory”

Material

https://github.com/gamati01/HPCLessons

You'll have access to

- ✓ PDF of the various presentations
- ✓ Code in Fortran/C
  - to complete
  - to modify
  - to simply run
- ✓ `git clone https://github.com/gamati01/HPCLessons.git`